

# BORAMA HARDWARE

## App Flow & User Journey Document

Complete Screen Map • Every State • Every Transition

Customer Store • Admin Portal • Inventory Manager • Staff POS

Version 1.0 • July 2025 • Confidential

| Field         | Details                                                 |
|---------------|---------------------------------------------------------|
| Document Type | App Flow & User Journey Document                        |
| Project       | Borama Hardware Store                                   |
| Version       | 1.0                                                     |
| Date          | July 2025                                               |
| Status        | Final — Ready for Development                           |
| Portals       | Customer Store / Admin / Inventory Manager / Staff POS  |
| Stack         | Next.js 14 (App Router) · Supabase · Vercel · Gemini AI |
| Total Screens | 70+ distinct screens and states documented              |

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# Part 1 — Navigation Architecture & Route Map

The application is a single Next.js 14 App Router monorepo serving four distinct portals under one domain. Each portal has its own layout, navigation chrome, and role-based access guard. All portals share the same Supabase backend and authentication session.

## 1.1 Global Route Structure

| Route Prefix | Portal            | Accessed By               | Layout Chrome                         |
|--------------|-------------------|---------------------------|---------------------------------------|
| /            | Customer Store    | Public + Customers        | Header + bottom nav (mobile) + footer |
| /auth/*      | Authentication    | All unauthenticated users | Minimal — logo only                   |
| /account/*   | Customer Account  | Customers (authenticated) | Header + back button                  |
| /admin/*     | Admin Portal      | Admin / Owner             | Sidebar (collapsible) + top bar       |
| /inventory/* | Inventory Manager | Inventory Manager + Admin | Sidebar + top bar                     |
| /pos/*       | Staff POS         | Cashier + Admin           | Full-screen, no nav chrome            |
| /staff/*     | Staff Portal      | All internal staff        | Top bar + role tabs                   |

## 1.2 Layout System

Each portal uses a Next.js layout.tsx file as the persistent chrome wrapper. Layouts are nested — the root layout applies global providers (React Query, Zustand, next-intl, Supabase session). Portal-specific layouts add their navigation chrome. Page components render inside the layout's {children} slot.

| Layout File                | Wraps                | Contains                                                   |
|----------------------------|----------------------|------------------------------------------------------------|
| app/layout.tsx             | Everything           | Providers, HTML shell, global fonts, PWA meta              |
| app/(store)/layout.tsx     | Customer store pages | Store header, language toggle, cart icon, chatbot widget   |
| app/(auth)/layout.tsx      | Auth pages           | Centered card container, brand logo                        |
| app/(admin)/layout.tsx     | Admin pages          | Sidebar nav, top bar with user menu, notification bell     |
| app/(inventory)/layout.tsx | Inventory pages      | Sidebar nav (inventory variant), location selector top bar |
| app/(pos)/layout.tsx       | POS pages            | Full-screen no-chrome wrapper, PIN lock overlay            |
| app/(staff)/layout.tsx     | Staff portal pages   | Top bar, role-based tab navigation                         |

## 1.3 Persistent UI Elements

| Element           | Present On              | Behaviour                                                                                          |
|-------------------|-------------------------|----------------------------------------------------------------------------------------------------|
| Orange header bar | Customer store          | Sticky top. Logo left, search center, language toggle + cart + account right. Collapses on mobile. |
| Bottom navigation | Customer store (mobile) | Home, Categories, Search, Orders, Account. Active tab = orange icon + label. Hidden on desktop.    |

| Element             | Present On        | Behaviour                                                                                     |
|---------------------|-------------------|-----------------------------------------------------------------------------------------------|
| Sidebar navigation  | Admin / Inventory | 256px wide. Collapsible to 64px icon-only rail. Active item = orange background. Badge        |
| Chatbot widget      | Customer store    | Floating orange circle button. Bottom-right. Opens as drawer on mobile, floating panel o      |
| Offline banner      | All portals       | Yellow sticky banner below header. 'You are offline — some features are limited.' Dism        |
| Toast notifications | All portals       | Top-right stack on desktop, bottom-center on mobile. Auto-dismiss 4s. Success=green,          |
| Language toggle     | Customer store    | EN / SO pill toggle in header. Switches all text instantly. Persists to user profile if logge |
| Connection status   | POS only          | ONLINE (green dot) / OFFLINE (red dot) badge in POS top bar. Persistent — never hic           |

## Part 2 — Authentication Flows

**Authentication method:** Phone number + SMS OTP for all user types. No passwords, no email. Supabase Auth handles OTP delivery via Twilio SMS. Somaliland phone prefix validation enforced before send: 063, 065, 068, 061, 062, 066.

### 2.1 New Customer Registration Flow

| Step | Screen           | Route            | User Action / System Response                                                                              |
|------|------------------|------------------|------------------------------------------------------------------------------------------------------------|
| 1    | Phone Entry      | /auth/login      | User enters phone number. Prefix validated in real-time. 'Send Code' button activated.                     |
| 2    | OTP Verification | /auth/verify     | 6-digit OTP input (auto-advance digits). 60-second countdown resend timer. 3 incorrect attempts → lockout. |
| 3    | Onboarding       | /auth/onboarding | First-time only. Full name field (required). Language preference selector (English / Somali).              |
| 4    | Customer Store   | /                | Redirect to homepage with 'Welcome!' toast. Cart from IndexedDB merged with server data.                   |

### 2.2 Returning Customer Login Flow

Same as registration steps 1–2. After OTP verification, onboarding screen is skipped. User is redirected to their originally-intended destination (stored in session before redirect to auth) or the homepage if no prior intent.

### 2.3 Staff / Admin Login Flow

| Step | Screen           | Route          | User Action / System Response                                                         |
|------|------------------|----------------|---------------------------------------------------------------------------------------|
| 1    | Phone Entry      | /auth/login    | Same phone entry screen as customer. Staff phone must exist in the staff table.       |
| 2    | OTP Verification | /auth/verify   | Same OTP screen. On success, role is checked from the profiles table.                 |
| 3    | Role Redirect    | —              | Admin → /admin/dashboard. Inventory Manager → /inventory/dashboard. Cashier → /pos.   |
| 4    | POS PIN Setup    | /pos/setup-pin | First-time POS cashier only. Set 4-digit PIN. Confirm PIN. PIN stored as bcrypt hash. |

### 2.4 Session Expiry & Re-Authentication

Supabase access tokens expire after 1 hour. The refresh token is stored in an httpOnly cookie and auto-renewed by the Supabase client on each API call. If the refresh token also expires (7-day inactivity), the user is signed out silently and redirected to /auth/login with the message 'Your session has expired. Please sign in again.' Any unsaved form data is preserved in sessionStorage and restored after re-login.

### 2.5 Change Phone Number Flow

| Step | Action                                                                |
|------|-----------------------------------------------------------------------|
| 1    | Customer navigates to Account Settings → Change Phone Number.         |
| 2    | Enters new phone number. Validation runs against Somaliland prefixes. |

| Step | Action                                                                      |
|------|-----------------------------------------------------------------------------|
| 3    | OTP sent to the NEW number. Not the existing number.                        |
| 4    | Customer enters OTP on the new phone. On success, phone number is updated.  |
| 5    | Existing session remains valid. Toast: 'Phone number updated successfully.' |

2.6 Authentication Error States

| Error                       | Trigger                                    | Recovery Action                                                                  |
|-----------------------------|--------------------------------------------|----------------------------------------------------------------------------------|
| Invalid phone format        | Non-Somaliland prefix or wrong digit count | Online red error under input. Button stays disabled.                             |
| OTP expired                 | 60 seconds elapsed                         | 'Resend Code' button activates. Sends new OTP. Old code invalidated.             |
| OTP incorrect (3× failed)   | Three wrong codes entered                  | Screen locked for 15 minutes. Countdown shown. Cannot request new code.          |
| Rate limit hit              | More than 3 OTP requests per hour          | Error: 'Too many requests. Try again in X minutes.'                              |
| Unregistered staff phone    | Staff phone not in staff table             | Error: 'This number is not registered as a staff account. Contact your manager.' |
| Network failure on OTP send | Database/Twilio unreachable                | 'Could not send code. Check connection and try again.' Retry button shown.       |

## Part 3 — Customer Store Flows

### 3.1 Homepage (/) & Navigation

The homepage is the primary discovery surface. It loads via SSR for fast FCP, then hydrates for interactivity. All sections have skeleton loading states during hydration.

| Section                  | Content                                                                                             | State if Empty                         |
|--------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------|
| Hero banner carousel     | Admin-configured promotional banners. Auto-advanced. Single static banner gradient with store name. | Section hidden entirely.               |
| Category grid            | Top 8 categories as icon tiles. 'View All Categories' link.                                         | Never empty — seeded at launch.        |
| Featured products        | Admin-marked featured products. Horizontal scroll or section.                                       | Section hidden entirely.               |
| New arrivals             | Products sorted by created_at DESC, last 14 days.                                                   | Section hidden entirely.               |
| AI Project Estimator CTA | Large orange banner — 'Describe your project, we'll do everything you need.'                        | Always showing.                        |
| Special offers           | Products with active discounts.                                                                     | Section hidden if no active discounts. |

**Offline homepage:** If the user is offline and the page was previously visited, service worker returns the cached page. A yellow offline banner appears. Banner sections that require live data (AI sections, featured) show a 'Content not available offline' placeholder. Categories and cached products from a previous session are still browsable.

### 3.2 Product Listing Page (/category/[slug])

| Element             | Behaviour                                                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Page header         | Category breadcrumb trail (Home > Hardware > Fasteners). Category icon and name (EN or SO based on language).                     |
| Filter sidebar      | Desktop: sticky left column. Mobile: slides up from bottom as a full-screen drawer triggered by 'Filters' button. Filter chips.   |
| Sort control        | Dropdown: Featured, Price: Low to High, Price: High to Low, Newest, Best Selling. Default: Featured.                              |
| Product grid        | 3 columns desktop, 2 columns mobile. 24 products per page. Numbered pagination at bottom. Each card: image, title, price, status. |
| Active filter chips | Shown above grid. Each applied filter shown as a dismissible chip. 'Clear All' removes all filters.                               |
| Result count        | 'Showing 24 of 156 products' label above grid.                                                                                    |
| URL state           | Filters and sort written to URL query params (?brand=hilti&sort=price_asc&page=2). Browser back button restores state.            |

**Empty listing state:** When no products match the active filters, show: large illustration, 'No products found' (EN) / 'Alaab lama helin' (SO), 'Try adjusting your filters' suggestion, and a 'Clear Filters' button. Never show a blank grid.

### 3.3 Product Detail Page (/product/[slug])



| Zone                  | Content & Behaviour                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------|
| Image gallery         | Primary image large. Thumbnail strip below (up to 8). Click thumbnail = swap primary. Pinch-to-zoom on mobile. S         |
| Product identity      | Brand name (small, grey). Product name (H1, EN or SO). SKU badge. Category breadcrumb.                                   |
| Pricing               | SLSH price (large, bold). USD equivalent (small, grey). If variant selected and variant has different price, price up    |
| Variant selector      | Each variant attribute (e.g., Size, Colour) shown as a labelled row of pill buttons. Selecting a variant combination     |
| Stock status          | Per-location stock badge: 'In Stock' (green), 'Low Stock' (amber, quantity shown), 'Out of Stock' (red). If multiple lo  |
| Quantity selector     | - / [number] / + stepper. Max = available quantity. Cannot exceed stock. Min = 1.                                        |
| Add to Cart button    | Disabled (greyed) until all variants selected and quantity > 0. On tap: button pulses orange, item added, cart coun      |
| Loyalty points        | 'Earn XXX points with this purchase' shown below price in gold colour with star icon.                                    |
| Project Estimator CTA | Card: 'Using this product for a project? Get a full materials list.' → links to /estimator with this product pre-filled. |
| AI Description        | Collapsible 'Product Details' section. Bilingual description. 'Read in Somali / Read in English' toggle.                 |
| Related products      | AI-powered semantic recommendations: 4 cards in horizontal scroll. Labelled 'Customers also buy'.                        |
| Share button          | WhatsApp share icon — opens WhatsApp with product URL pre-filled.                                                        |

3.4 Search (/search?q=...)

| State                 | Behaviour                                                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------|
| Search input          | Full-width at top of page. Auto-focused on arrival. Keyboard shown on mobile immediately. Previous searches sho         |
| Instant suggestions   | After 300ms debounce, dropdown shows: matched product names (up to 5) with category, and a 'Search for [quer            |
| AI semantic search    | On search submit: Gemini text-embedding-004 converts query to vector, matches against pgvector embeddings in            |
| Results page          | Grid identical to category listing page. Result count: 'X results for [query]'. Bilingual results — searching in Somali |
| No results state      | Illustration + 'No results for [query]' + 'Try searching in Somali/English' + 3 suggested popular categories.           |
| Offline search        | Falls back to cached full-text search on locally stored product catalog. Results clearly labelled 'Results from your l  |
| Search result filters | Same filter sidebar as category listing. Filters apply on top of search results.                                        |

3.5 Cart — Mini-Cart Drawer & Full Cart (/cart)

**Mini-Cart Drawer** — Opens from the right on add-to-cart or cart icon tap. Does not navigate away from current page.

| Element           | Behaviour                                                                                                                   |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Drawer header     | Cart icon + 'My Cart' label + item count. X button closes. Closes also on Escape key or clicking the scrim.                 |
| Item list         | Product image, name, variant, SLSH + USD price, quantity stepper, remove (x) icon. Max 5 visible before scroll.             |
| Subtotal          | Running subtotal in SLSH. 'View Cart' and 'Checkout' buttons. If cart empty: illustration + 'Your cart is empty' + 'Start s |
| Offline indicator | If offline: 'Checkout not available offline' message. Checkout button disabled. Items still editable.                       |

Full Cart Page (/cart)

| Element             | Behaviour                                                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Cart items          | Same controls as mini-cart. Inline quantity update triggers instant price recalculation without page reload.                  |
| Discount code       | Text input + 'Apply' button. On success: discount amount shown in green, 'Remove' link. On invalid: red error 'Invalid'       |
| Loyalty redemption  | If logged in and has points: 'Redeem X points for Y% off' toggle. Toggle on = discount applied. Mutually exclusive with       |
| Location selector   | Dropdown: which store branch to pick up from. Required before checkout. Stock validated against selected location.            |
| Order summary       | Subtotal, discount (if applied), loyalty discount (if applied), delivery (always 'Free — in-store pickup'), total in SLSH +   |
| Proceed to Checkout | Button at bottom right. Disabled if: no location selected, any item out of stock at selected location. Stock is re-validated  |
| Stock conflict      | If re-validation finds an item out of stock, that item's card turns red with 'Out of stock at [location]'. Remove to continue |

3.6 Checkout (/checkout)

Single-page checkout. All sections visible at once on desktop. On mobile, sections stack vertically with the order summary collapsible at top.

| Section            | Fields & Behaviour                                                                                                 |
|--------------------|--------------------------------------------------------------------------------------------------------------------|
| Pickup location    | Read-only — shows the location selected in cart. 'Change' link takes back to cart.                                 |
| Contact details    | Phone number (pre-filled if logged in, read-only). Full name (pre-filled, editable).                               |
| Payment method     | Radio group: Zaad / eDahab / EVC Plus / Sahal / Cash on Pickup. One selected at a time.                            |
| Mobile money phone | Shown only if mobile money selected. 'Which number did you send from?' — for staff reference. Validates Somali     |
| Order notes        | Optional textarea. Placeholder: 'Any special instructions for pickup?' Max 500 chars.                              |
| Order summary      | Collapsible on mobile. Shows all items with quantities, discount applied, totals.                                  |
| Place Order button | Primary orange button. Triggers loading spinner. Disabled during processing. On success: redirect to /order-confir |
| Error on submit    | If API fails: 'Could not place your order. Please try again.' Toast. Form stays populated. Try Again button.       |

**Guest checkout:** Not supported. Unauthenticated users clicking 'Checkout' are redirected to /auth/login with return URL /checkout. Cart is preserved in IndexedDB through the auth flow and merged back on login.

3.7 Order Confirmation (/order-confirmation/[id])

| Element              | Content                                                                                                          |
|----------------------|------------------------------------------------------------------------------------------------------------------|
| Success animation    | Green checkmark animation plays once on page load (respects prefers-reduced-motion — shows static check ins      |
| Order number         | Large bold. 'Order #BH-2025-00042'                                                                               |
| Pickup code          | 6-character alphanumeric code in very large font (e.g., XK7R2P). Primary visual element for cash on pickup order |
| Payment instructions | Shown only for mobile money orders. Admin-configured bilingual instructions: 'Send [AMOUNT] SLSH to Zaad nu      |

| Element               | Content                                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------------------|
| 'I Have Sent Payment' | Shown only for mobile money. Large button. Tap = sets payment status to pending_verification. Button disappears |
| Payment confirmed     | When staff confirms via Supabase Realtime: spinner replaced by green 'Payment Confirmed!' with checkmark and    |
| WhatsApp share        | Button: 'Share with someone collecting for you.' Opens WhatsApp with pre-written message: order number, pickup  |
| 'View My Orders' link | Always visible at bottom.                                                                                       |
| Back button behaviour | Browser back from this page goes to homepage — not to checkout. Prevents accidental re-submission.              |

3.8 Order History, Detail & Cancellation

/account/orders — Order List

| Element       | Behaviour                                                                                                              |
|---------------|------------------------------------------------------------------------------------------------------------------------|
| Order list    | Cards sorted newest-first. Each: order number, date, item count thumbnail strip (max 3 thumbnails + '+X more'), total, |
| Status badges | Pending Verification (amber), Confirmed (blue), Ready for Pickup (purple), Completed (green), Cancelled (grey), Retu   |
| Empty state   | Illustration + 'No orders yet' (EN) / 'Dalabyo ma jiraan' (SO) + 'Start Shopping' button.                              |
| Pagination    | Load More button at bottom (loads 10 more). Not infinite scroll — intentional to aid navigation.                       |

/account/orders/[id] — Order Detail

| Element           | Behaviour                                                                                                            |
|-------------------|----------------------------------------------------------------------------------------------------------------------|
| Status timeline   | Visual step-by-step tracker: Placed → Confirmed → Ready for Pickup → Completed. Current step highlighted in oran     |
| Pickup code       | Shown prominently if order is Confirmed or Ready for Pickup. Hidden after Completed.                                 |
| Real-time updates | Page subscribes to Supabase Realtime on the order record. Status changes update live without refresh.                |
| Cancel order      | Shown if status is Pending or Pending Verification. Button opens confirmation modal: 'Are you sure? Select a reason: |
| Return items      | Shown if status is Completed and within 7 days of completion. 'Return Items' button → /account/returns/new?order=[i  |
| Reorder           | Button at bottom: adds all items from this order back to cart (validates stock, alerts on any out-of-stock items).   |

3.9 Loyalty Program (/account/loyalty)

| Element             | Content & Behaviour                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------|
| Loyalty card        | Animated orange card with customer name, unique loyalty card number (barcode), current points balance, curren      |
| Tier progress bar   | Visual progress to next tier. Shows points needed. Example: 'You need 230 more points for Gold status.' If already |
| Tier benefits       | Table comparing Bronze / Silver / Gold benefits side by side.                                                      |
| Transaction history | List of all point events: earned (green +), redeemed (red -), adjusted (grey ±). Infinite scroll.                  |
| Tier upgrade modal  | When a purchase causes a tier upgrade: fullscreen confetti animation plays, modal: 'You reached [Tier]! 🎉' with n  |

### 3.10 AI Project Estimator (/estimator)

| Step  | Screen State | Behaviour                                                                                                                                                                                                     |
|-------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Input        | Large textarea: 'Describe your project in English or Somali.' Examples shown as clickable chips: 'Tile a 3m x 3m area', 'Estimate materials for a kitchen renovation', 'Calculate cost for a room extension'. |
| 2     | Processing   | Full-screen loading state: pulsing orange logo + 'Estimating materials...' + 'Analysing your project' sub-text. A progress bar indicates the status.                                                          |
| 3     | Results      | Titled section per category (e.g., Tiles, Adhesive, Grout). Each item: product name (linked to detail page), quantity, price per unit, and total price for the project.                                       |
| 4     | Add to Cart  | 'Add All to Cart' button adds all in-stock items. Out-of-stock items shown with 'Not available' badge — excluded from total.                                                                                  |
| 5     | Refine       | 'Adjust Project Details' button returns to input with previous values pre-filled for editing.                                                                                                                 |
| Error | AI failure   | If Gemini fails: 'Could not generate estimate. Please try again.' with retry. If project description too vague: 'Your description is too vague. Please be more specific.'                                     |

### 3.11 Hardware Assistant Chatbot

Floating orange chat icon, bottom-right of customer store on all pages. Opens as a drawer on mobile (80% screen height), floating card on desktop (400×600px). Powered by Gemini 1.5 Flash. Rate limit: 20 messages/hour per user.

| State            | Behaviour                                                                                                                                                                              |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Welcome state    | On first open: 'Hi! I'm your hardware assistant. Ask me anything about tools, materials, or your project.' 3 quick-action buttons: 'Get Recommendations', 'Check Cart', 'Track Order'. |
| Typing indicator | Animated three-dot pulsing while AI generates response. Cannot send next message while waiting.                                                                                        |
| Product links    | When AI recommends a product from the catalog, it includes a clickable product card (thumbnail, name, price) instead of just text.                                                     |
| Language         | Detects language of user's message automatically. Responds in the same language (English or Somali).                                                                                   |
| History          | Conversation persists for the session. Cleared on logout or manual 'Clear chat' button.                                                                                                |
| Rate limit hit   | 'You've reached the hourly message limit. Try again in X minutes.' Input disabled until limit resets.                                                                                  |
| Offline          | Chatbot icon greyed out with offline indicator. Tap shows: 'Chat is not available offline.'                                                                                            |

## Part 4 — Admin Portal Flows (/admin/\*)

**Access:** Admin and Owner roles only. Supabase RLS enforces all data access. Admins see all locations. Admin sidebar is always visible on desktop. On mobile (tablet), sidebar collapses to icon-only rail — tap icon to expand as overlay.

### 4.1 Admin Dashboard (/admin/dashboard)

| Section                  | Content & Behaviour                                                                                                |
|--------------------------|--------------------------------------------------------------------------------------------------------------------|
| Stat cards (row 1)       | Today's Revenue (SLSH), Orders Today, New Customers Today, Active Alerts. Each card: value, % change vs y          |
| Exchange rate alert      | Amber banner if rate hasn't been updated in 24+ hours: 'Exchange rate is X hours old. Update now.' Dismissible b   |
| Revenue chart            | Line chart — last 30 days, one point per day. Toggle: This Month / Last Month / Last 3 Months. Filterable by locat |
| Top products table       | Top 10 by revenue in selected period. Columns: product name, units sold, revenue. Clickable row → product deta     |
| Orders by location       | Horizontal bar chart. One bar per location.                                                                        |
| Orders by payment method | Donut chart: Zaad / eDahab / EVC Plus / Sahal / Cash. Percentages in legend.                                       |
| First-time setup banner  | On first login with no products: setup checklist (see 4.1.1). Dismissed permanently once all steps completed.      |

#### 4.1.1 First-Time Setup Checklist

| # | Step                    | Route                    | Completion Trigger                     |
|---|-------------------------|--------------------------|----------------------------------------|
| 1 | Set exchange rate       | /admin/settings/exchange | Exchange rate saved for first time     |
| 2 | Add your first location | /admin/locations/new     | First location record created          |
| 3 | Add payment numbers     | /admin/settings/payments | At least one mobile money number saved |
| 4 | Add product categories  | /admin/categories        | At least one category created          |
| 5 | Add your first product  | /admin/products/new      | First product published                |
| 6 | Add staff members       | /admin/staff             | At least one staff account added       |

### 4.2 Product Management (/admin/products)

| Screen           | Behaviour                                                                                                          |
|------------------|--------------------------------------------------------------------------------------------------------------------|
| Product list     | Paginated table (25/page). Columns: image thumbnail, name, SKU, category, SLSH price, stock (total across loca     |
| New product form | Tabbed form: Basic Info   Pricing   Images   Variants   Inventory   SEO. Must complete Basic Info before other tab |
| Basic Info tab   | Product name (EN + SO side by side). Category selector (hierarchical). Brand. Unit (piece, kg, m, box). Tags. 'Ge  |
| Pricing tab      | USD price (primary input). SLSH auto-calculated at current exchange rate (editable override). Cost price (SLSH) f  |
| Images tab       | Drag-and-drop upload. Up to 8 images. Drag to reorder (first = primary). Each image: preview, remove button, alt   |

| Screen                | Behaviour                                                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Variants tab          | Variant attribute groups. 'Add Attribute' button opens modal: Attribute name (e.g., Size), Attribute values (comma-separated)   |
| Inventory tab         | Per-location quantity on hand. Low stock threshold per product. Aisle/shelf location (text). These are set per brand            |
| AI Description button | Sends product name, category, brand to Gemini Flash. Response pre-fills EN and SO description fields. Spinner on response       |
| Save & Publish        | Validates all required fields. If images missing: warning (not error) — can save without images. On success: toast message      |
| Bulk CSV Import       | Button: 'Import Products'. Modal with: download template CSV button, file upload zone, preview of first 5 rows, 'Import' button |

4.3 Order Management (/admin/orders)

| Screen               | Behaviour                                                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Order list           | Table with filters: Location, Status, Payment Method, Date Range (datepicker). Export filtered list to CSV button. Search bar |
| Order detail         | Header: order number, date, customer info, payment status. Items table. Payment section: method, reference number, amount     |
| Payment confirmation | Admin enters transaction reference from store phone. On confirm: order status → Confirmed, SMS sent to customer               |
| Cancel order         | Modal: confirm cancel, reason required (dropdown), optional note. Cannot cancel orders that are Completed or have shipped     |
| Initiate return      | Admin can initiate a return on behalf of a customer from the order detail. Same form as customer-initiated return.            |

4.4 Staff, Location & Settings Management

| Section                                    | Key Behaviours                                                                                                                   |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Staff (/admin/staff)                       | Table: name, phone, role badge, location, status. 'Add Staff' modal: phone number input (validates), role selector, save button  |
| Locations (/admin/locations)               | Table: name, address, phone, hours. 'Edit' opens form: name (EN+SO), address, phone, hours per day (dropdown)                    |
| Discounts (/admin/discounts)               | Table with code, type, value, usage count vs limit, validity, status. 'Add Discount' form: code (auto-generate or manual), value |
| Exchange Rate (/admin/exchange-rate)       | Single form to update SLSH rate input. Save button. History table: 10 most recent rates, who set them, and when.                 |
| Payment Settings (/admin/payment-settings) | Settings provided by Zaad, eDahab, EVC Plus, Sahal). Each tab: merchant number per location, custom payment instructions         |
| Banner Management (/admin/banners)         | Drag-and-drop ordered list of banners. Each: image upload (1200×400px recommended), title (EN+SO), CTA text (EN+SO)              |
| Loyalty Config (/admin/loyalty)            | Tier threshold inputs: Silver (default 500), Gold (default 2000). Tier discount percentages. Tier names (EN+SO). Save button     |

4.5 AI Insights & Analytics (/admin/insights)

| Section             | Behaviour                                                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Inventory Forecasts | Table: Product, Location, Current Stock, Predicted Stockout Date, Recommended Reorder Qty, Confidence Score                   |
| Forecast detail     | Click any row → side panel with AI reasoning text, 30/60/90-day sales chart for that product, comparable seasonal trends      |
| Analytics Reports   | Revenue by period/location, top products, return rate trends, POS session summaries, staff performance table. Download button |

| Section                  | Behaviour                                                                                                                              |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Audit Log (/admin/audit) | Immutable table. Filters: date range, user, table name. Columns: timestamp, user, table, record ID, action (created, updated, deleted) |

## Part 5 — Inventory Manager Flows (/inventory/\*)

**Location scope:** Inventory Managers see only their assigned location's data. Admins can switch location from the top bar dropdown to view any branch.

### 5.1 Inventory Dashboard (/inventory/dashboard)

| Card / Section       | Content                                                                                                                  |
|----------------------|--------------------------------------------------------------------------------------------------------------------------|
| Summary cards        | Total SKUs tracked, Items in Stock (sum of all quantities), Low Stock count (items below threshold), Out of Stock count  |
| Alert feed           | Live list of unresolved alerts sorted by severity. Each: product name, variant, current qty, threshold, time since alert |
| Stock overview chart | Horizontal bar chart: top 10 products by stock quantity. Filtered to current location.                                   |
| Recent movements     | Last 10 stock movements: type (receive/adjust/transfer/sale), product, quantity change, user, time.                      |

### 5.2 Stock Levels (/inventory/stock)

| Element           | Behaviour                                                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Filters           | Search by product name or SKU. Filter by category, status (In Stock / Low Stock / Out of Stock). Sort by: name, quantity            |
| Stock table       | Columns: Product (image + name), Variant, SKU, On Hand, Reserved (orders pending pickup), Available (on hand minus reserved)        |
| Inline adjustment | Click quantity cell → inline editable number. Enter = save. Creates a stock_movement record (type: adjustment) with quantity change |
| Quick receive     | From stock table row → 'Receive Stock' opens pre-filled form for that product/variant.                                              |
| Export            | Export current filtered view to CSV.                                                                                                |

### 5.3 Stocktake Full Workflow (/inventory/stocktake)

| Step | Screen              | Behaviour                                                                                                                                 |
|------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Initiate Stocktake  | Button: 'Start New Stocktake'. Confirmation modal: 'This will lock stock adjustments for [location] until the stocktake is completed.'    |
| 2    | Count Sheet         | Full table of all active products/variants at this location. Columns: Product, Variant, SKU, System Qty (frozen), Counted Qty, Difference |
| 3    | Barcode scanning    | Focus on Counted Qty input for a product by scanning its barcode. Scanner input detected by timing (200ms)                                |
| 4    | Save progress       | Auto-saves every 2 minutes to prevent data loss. 'Save Draft' button. Can leave and return — count sheet persists                         |
| 5    | Discrepancy review  | After all items counted, 'Review Discrepancies' button shows only items with non-zero difference. Staff can adjust quantities             |
| 6    | Submit for approval | 'Submit Stocktake' → status: pending_approval. Inventory Manager cannot edit after submission. Email/SMS to admin                         |
| 7    | Admin approval      | Admin sees stocktake in /admin/stocktakes. Reviews discrepancies. Approves = all counted quantities become official                       |
| 8    | Completion          | Status: completed. Audit log entry created. Stock lock released.                                                                          |



## 5.4 Receive Stock (/inventory/receive)

| Field        | Behaviour                                                                                                             |
|--------------|-----------------------------------------------------------------------------------------------------------------------|
| Location     | Locked to user's assigned location (Inventory Manager) or selectable dropdown (Admin).                                |
| Product rows | Dynamic form rows. Each row: product search/select (autocomplete), variant selector (appears if product has variants) |
| Submit       | Creates stock_movement records (type: receive) for each row. Increments inventory.quantity_on_hand. Updates last      |
| Success      | Toast: 'Stock received successfully. X products updated.' Form resets.                                                |

## Part 6 — Staff / POS Flows (/pos/\* & /staff/\*)

**POS design principle:** The POS is full-screen with no browser chrome. It runs in a dedicated layout with no external navigation. All interactions are optimised for touch on a countertop tablet. Connectivity state is always visible. Offline cash sales always continue.

### 6.1 POS Session Open (/pos/open-session)

| Step | Element              | Behaviour                                                                                                       |
|------|----------------------|-----------------------------------------------------------------------------------------------------------------|
| 1    | PIN login            | Cashier arrives at POS, enters 4-digit PIN. 3 failures = full OTP re-auth required.                             |
| 2    | Open Session form    | Starting cash: SLSH input + USD input (both optional but encouraged). Location confirmed (locked to cashier).   |
| 3    | POS Interface        | Navigates to /pos/session. Session record created with start time, starting cash, and cashier ID.               |
| 4    | Session already open | If a session is already open (cashier forgot to close yesterday), shown: 'Session from [date] is still open. Co |

### 6.2 POS Main Interface (/pos/session)

Dual-column layout. Left panel: product search + category grid. Right panel: cart + payment. No scrolling on the main view — everything visible simultaneously.

| Panel          | Element               | Behaviour                                                                                                |
|----------------|-----------------------|----------------------------------------------------------------------------------------------------------|
| Left — Top     | Search bar            | Auto-focused. Type product name or SKU. Matches update product grid below in real time. Clear b          |
| Left — Top     | Category quick-filter | Row of category pill buttons below search. Tap = filters product grid to that category. Active = oran    |
| Left — Main    | Product grid          | Large touch-friendly tiles (3 columns). Each: product image, name, SLSH price, stock count. Tap =        |
| Left — Top     | Barcode input         | Invisible input permanently focused in background. Barcode scan detected by 200ms keystroke tim          |
| Right — Top    | Cart header           | Transaction #, cashier name, date. 'Park' button (holds cart). 'Parked' button (recalls parked carts l   |
| Right — Middle | Cart items            | Product, variant, qty stepper, unit price, line total, remove (×). Max 15 items visible before scroll. T |
| Right — Bottom | Cart footer           | Discount code input. Subtotal. Tax (0% — shown but zero for Somaliland). Total in SLSH (large). '        |
| Right — Bottom | Customer lookup       | Phone input field: 'Link customer for loyalty points'. Lookup = live search. Matched customer show       |
| Top bar        | Connectivity badge    | ONLINE (green dot) or OFFLINE (red dot). Session timer. Location name. Cashier name. PIN lock            |

### 6.3 Payment Flows

#### Cash Payment

| Step | Behaviour                                                       |
|------|-----------------------------------------------------------------|
| 1    | 'Charge' tapped → payment modal opens. Total shown prominently. |

| Step | Behaviour                                                                                                                  |
|------|----------------------------------------------------------------------------------------------------------------------------|
| 2    | Payment method: Cash selected (default). 'Amount Tendered' field.                                                          |
| 3    | As cashier types received amount, 'Change Due' calculates and displays live (green if positive, red if insufficient).      |
| 4    | 'Complete Sale' button active only when tendered $\geq$ total. Tap = order created (status: completed). Receipt generated. |
| 5    | Receipt screen: print button, download PDF button, 'New Sale' button (clears cart).                                        |

### Mobile Money Payment (POS)

| Step | Behaviour                                                                                                                     |
|------|-------------------------------------------------------------------------------------------------------------------------------|
| 1    | Payment method: select Zaad / eDahab / EVC Plus / Sahal. Order created (status: pending_verification).                        |
| 2    | Screen shows: provider merchant number for this location, exact amount, order number as reference. Cashier reads to customer. |
| 3    | Customer pays from their own phone. Cashier receives incoming transfer notification on store phone.                           |
| 4    | Cashier enters transaction reference from store phone. 'Confirm Payment' button.                                              |
| 5    | Order status $\rightarrow$ completed. Receipt generated. Customer SMS sent.                                                   |

### Split Payment

| Step | Behaviour                                                                                                                              |
|------|----------------------------------------------------------------------------------------------------------------------------------------|
| 1    | In payment modal, toggle 'Split Payment' switch.                                                                                       |
| 2    | Split builder: Add payment method rows. First row pre-filled with the full total. Adding a second row prompts for amount — remaining b |
| 3    | Each method has its own amount field. Total of all splits must equal order total — validation prevents completion otherwise.           |
| 4    | Cash splits calculate change within their own row. Mobile money splits each create a separate verification record.                     |
| 5    | All splits confirmed $\rightarrow$ order completed. Receipt shows all payment methods and amounts.                                     |

### Void Transaction

| Step | Behaviour                                                                                                                        |
|------|----------------------------------------------------------------------------------------------------------------------------------|
| 1    | 'Void' button accessible from the session transaction history list (within the current session only, within 60 minutes of sale). |
| 2    | Confirmation modal: 'Void transaction #BH-POS-0042? This cannot be undone. Reason required.'                                     |
| 3    | On confirm: order voided, inventory restored, void_transaction record created, session refund total incremented.                 |
| 4    | Void receipt generated with large 'VOID' watermark. Print/download options.                                                      |
| 5    | Cannot void mobile money orders where payment has already been confirmed — must initiate return instead.                         |

## 6.4 PIN Lock Screen (/pos/lock)

POS locks after 5 minutes of inactivity (no cart interaction, no taps on screen). Current cart state is preserved. Lock overlay covers the entire screen.

| Element      | Behaviour                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------|
| Lock display | Cashier name + 'POS Locked' message. Clock showing current time. Location name.                                |
| PIN pad      | 4-digit dot display. Number pad 0–9 + delete. Enter PIN to unlock.                                             |
| 3 failures   | 'Too many attempts. Please re-authenticate with your phone number.' → OTP flow. On OTP success, POS unlocks a  |
| Manual lock  | Cashier can manually lock from the top bar lock icon (e.g., going to assist a customer away from the counter). |

6.5 Close Session

| Step | Element            | Behaviour                                                                                                 |
|------|--------------------|-----------------------------------------------------------------------------------------------------------|
| 1    | Close button       | 'Close Session' in POS top bar. Confirmation: 'Are you sure? Parked transactions will be lost.'           |
| 2    | End cash count     | Fields: counted SL\$H cash, counted USD cash. System shows expected cash from opening + cash sales -      |
| 3    | Variance           | If counted cash differs from expected: variance shown in amber/red. 'Over/Short by X SL\$H' — cashier mus |
| 4    | Session summary    | Read-only summary: session duration, total sales, breakdown by payment method, total void amount, cash    |
| 5    | Close confirmation | 'Close Session' final confirm button. Session status → closed. Cash settlement record created.            |
| 6    | Print/export       | Option to print or download session summary as PDF.                                                       |

6.6 Order Pickup Management (/staff/pickup)

| Screen             | Behaviour                                                                                                           |
|--------------------|---------------------------------------------------------------------------------------------------------------------|
| Pickup queue       | Real-time list (Supabase Realtime) of all Confirmed orders for this location. Sorted by time placed. Cards: order n |
| 'Ready for Pickup' | Button on each card. Tap = order status → ready_for_pickup. Triggers SMS to customer: 'Your order #BH-2025-0        |
| Verify pickup code | Customer arrives. Staff taps 'Verify Pickup'. Modal: numeric keypad to enter customer's 6-character pickup code (   |
| Completed orders   | Tab: Completed Today. Shows all completions this session.                                                           |

## Part 7 — Returns & Refunds Flows

### 7.1 Customer — Initiate Return (/account/returns/new?order=[id])

| Step | Screen Element    | Behaviour                                                                                                                              |
|------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Select items      | Checklist of items from the order. Cannot select items already returned. Per item: checkbox, quantity to return.                       |
| 2    | Select reason     | Dropdown per item (or single reason for all): Wrong item delivered / Defective / Product not as described / Other.                     |
| 3    | Photos (optional) | Upload up to 3 photos as evidence. Compressed on device before upload. Max 5MB per image.                                              |
| 4    | Refund method     | Radio: Cash refund at store / Mobile money refund (enter phone number) / Store credit (loyalty points). Default: Cash refund at store. |
| 5    | Submit            | Return request created (status: pending). Customer sees: 'Return request submitted. We will contact you within 24 hours.'              |
| 6    | Return status     | Visible in order detail page under 'Return Requests'. Status: Pending → Approved → Refunded / Rejected.                                |

### 7.2 Staff — Process Return (/staff/returns)

| Step | Screen Element        | Behaviour                                                                                                             |
|------|-----------------------|-----------------------------------------------------------------------------------------------------------------------|
| 1    | Return queue          | List of pending returns for this location. Cards: order #, customer, items requested, reason, photos (expandable).    |
| 2    | Review                | Expand each item. View submitted photos. Condition inspection notes field (internal).                                 |
| 3    | Approve or Reject     | Approve: stock restored, loyalty points adjusted (if applicable), refund processed. Reject: require reason, SMS sent. |
| 4    | Mobile money refund   | If refund method = mobile money: entry field for transaction reference number (staff sends money from store account). |
| 5    | Cash refund           | Approve = generate 'Refund Due' receipt. Physical cash given at counter. Staff marks as 'Cash Handed Over'.           |
| 6    | Store credit refund   | Approve = X points added to customer's loyalty account. Points transaction record created.                            |
| 7    | Customer notification | On any outcome (approve/reject), SMS sent to customer automatically.                                                  |

## Part 8 — Cross-Portal & Real-Time Flows

### 8.1 Real-Time Data Flows

Supabase Realtime (PostgreSQL LISTEN/NOTIFY via websocket) powers all live updates. Each portal subscribes to the tables relevant to its role.

| Event                     | Source               | Subscribers                            | UI Effect                                                            |
|---------------------------|----------------------|----------------------------------------|----------------------------------------------------------------------|
| Order payment confirmed   | orders table update  | Customer order detail page             | Spinner → green 'Payment Confirmed' animation. No page refresh.      |
| Order status changed      | orders table update  | Customer order detail, Staff dashboard | Customer: status timeline updates. Staff: card moves between queues. |
| New online order          | orders table insert  | Staff pickup queue (/staff/pickup)     | New card animates in at top of queue with chime sound.               |
| New payment pending       | orders table insert  | Staff payment verification screen      | New card appears in unverified list.                                 |
| Low stock alert triggered | alerts table insert  | Inventory dashboard, Admin dashboard   | Alert badge increments, alert card animates into feed.               |
| New return request        | returns table insert | Staff returns, Admin returns           | Count badge on Returns nav item increments.                          |
| Stocktake submitted       | stocktakes table     | Admin                                  | Notification bell: 'Stocktake awaiting approval at [location]'.      |

### 8.2 Offline → Online Transition

The service worker (next-pwa) and Background Sync API handle reconnection. The following applies across all portals:

| Scenario                      | Offline Behaviour                                    | On Reconnect                                                                          |
|-------------------------------|------------------------------------------------------|---------------------------------------------------------------------------------------|
| Customer browsing             | Cached pages served. 'Offline' banner shown.         | Banner auto-hides. Live product data refreshed in background.                         |
| Customer places order offline | Checkout button disabled. Cart preserved.            | Checkout becomes available immediately. No auto-submission.                           |
| POS cash sale offline         | Sale processed, stored in IndexedDB queue.           | Background Sync fires. All queued sales POSTed to API in order. Sync indicator shows. |
| POS mobile money offline      | Mobile money options hidden with explanation.        | Options reappear automatically. No queued MM orders.                                  |
| Inventory view offline        | Last cached stock data displayed with stale warning. | Database refreshes automatically.                                                     |
| Admin offline                 | Read-only cached data. Save/edit actions disabled.   | Full functionality restored. Unsaved form data (in sessionStorage) preserved.         |

### 8.3 SMS Notification Flow

| Trigger Event           | Recipient | Message Content (EN, Somali version sent based on customer preference)                                      |
|-------------------------|-----------|-------------------------------------------------------------------------------------------------------------|
| Order placed            | Customer  | 'Your order #BH-0042 has been received. Total: 145,000 SLSH. Pickup location: [branch]'                     |
| Payment confirmed       | Customer  | 'Payment confirmed for order #BH-0042. Your pickup code is XK7R2P. We'll notify you when ready.'            |
| Order ready for pickup  | Customer  | 'Your order is ready! Show code XK7R2P at [branch] to collect your items.'                                  |
| Order cancelled         | Customer  | 'Order #BH-0042 has been cancelled. Contact us if you need help.'                                           |
| Return request approved | Customer  | 'Your return for order #BH-0042 has been approved. Refund of 45,000 SLSH will be credited to your account.' |

| Trigger Event              | Recipient   | Message Content (EN, Somali version sent based on customer preference)                             |
|----------------------------|-------------|----------------------------------------------------------------------------------------------------|
| Return request rejected    | Customer    | 'Your return for order #BH-0042 was not approved. Reason: [reason]. Visit us at [branch address].' |
| Low stock alert (internal) | Admin/Owner | 'Stock alert: [Product] at [branch] is below threshold. Current qty: [X].'                         |

**SMS failure handling:** Failed SMS deliveries (Twilio error) are logged to the sms\_logs table with the error code and retried automatically once. If the retry also fails, the operation continues normally — SMS delivery failure never blocks the order flow. Staff can manually resend from the order detail page.

## Part 9 — Complete Screen State Matrix

Every screen in the application has at least three distinct states: loading, populated, and empty/error. This matrix documents the complete set of states for all major screens.

### 9.1 Loading States

| Screen                | Loading Pattern                                                                       | Duration Target                  |
|-----------------------|---------------------------------------------------------------------------------------|----------------------------------|
| Homepage              | 3 skeleton banner blocks + category grid of 8 grey tiles + 4 product card skeletons   | < 1s                             |
| Product listing       | 12 product card skeletons in grid (same aspect ratio as real cards)                   | < 1.5s                           |
| Product detail        | Split skeleton: left=image placeholder, right=text lines of varying widths            | < 1s                             |
| Search results        | Search bar active, 'Searching...' text, 8 card skeletons below for AI semantic search | < 2s                             |
| Cart                  | 3 item skeletons, subtotal skeleton, grey button                                      | < 500ms                          |
| Checkout              | Form field skeletons, grey Place Order button                                         | < 500ms                          |
| Order list            | 5 order card skeletons with status badge placeholder                                  | < 1s                             |
| Admin dashboard       | Stat card skeletons, chart placeholder rectangles                                     | < 2s                             |
| POS session open      | Full-screen spinner with Borama Hardware logo centered                                | < 1.5s                           |
| AI Project Estimator  | Pulsing orange logo + 'Estimating materials...' animation                             | 3–5s (AI generation)             |
| AI Forecasts          | Table skeleton + 'Generating forecasts...' progress message                           | 15–30s (shown with progress bar) |
| Stocktake count sheet | Table skeleton row per product (shimmer animation)                                    | < 2s                             |

### 9.2 Empty States

All empty states include: an illustration, a title in the current language, a one-sentence explanation, and a primary action CTA button.

| Screen                    | English Text                        | Somali Text                      | CTA                 |
|---------------------------|-------------------------------------|----------------------------------|---------------------|
| Customer cart (empty)     | 'Your cart is empty'                | 'Baaskiidkaagu waa madhan yahay' | Start Shopping      |
| Order history (no orders) | 'No orders yet'                     | 'Dalabyo ma jiraan'              | Browse Products     |
| Search (no results)       | 'No results for [query]'            | 'Wax lama helin [query]'         | Browse Categories   |
| Loyalty history (no txns) | 'No transactions yet'               | 'Xisaab-gal ma jiraan'           | Shop to Earn Points |
| Admin orders (no orders)  | 'No orders match your filters'      | 'Dalabyo lama helin'             | Clear Filters       |
| Admin products (no prods) | 'No products yet — add your first!' | 'Alaab ma jiraan'                | Add Product         |
| Inventory alerts (none)   | 'All stock levels are healthy'      | 'Kaydku waa hagaagsan yahay'     | View Stock Levels   |
| POS — no results (search) | 'No products found'                 | 'Alaab lama helin'               | Clear Search        |



| Screen                    | English Text                                                | Somali Text                           | CTA                |
|---------------------------|-------------------------------------------------------------|---------------------------------------|--------------------|
| Staff pickup (no orders)  | 'No orders waiting for pickup'                              | 'Dalabyo sugaya ma jiraan'            | —                  |
| Staff returns (none)      | 'No return requests'                                        | 'Codsi celin ah ma jiraan'            | —                  |
| Chatbot (fresh open)      | 'Hi! Ask me anything about hardware'                        | 'Salaaan! I wax ka weydii qaladaabka' | Quick-action chips |
| Admin customers (none)    | 'No customers registered yet'                               | 'Macaamiil ma jiraan'                 | —                  |
| Admin analytics (no data) | 'Not enough data yet. Data appears after your first sales.' | 'Xagta ugu horeysa ma jirto.'         | —                  |

9.3 Error States

| Error Type                 | Trigger                                           | Display                                                              | Recovery                                           |
|----------------------------|---------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------|
| Network / API failure      | fetch() throws or 5xx response                    | Toast: 'Something went wrong. Try again later.'                      | Retry button. Auto-retry once after 2s.            |
| Stock conflict (checkout)  | Item out of stock at selected location            | Affected item card turns red with message.                           | Remove item or change location.                    |
| Payment timeout            | No staff confirmation within 30s                  | Customer sees: 'Your payment has timed out. Contact the staff here.' | Continue to counter or contact the staff. SMS adv. |
| Session expired            | Auth token cannot be refreshed                    | Redirect to /auth/login with toast: 'Session expired.'               | Re-login. Data restored.                           |
| POS offline MM attempt     | Mobile money selected while offline               | Payment method row shows lock icon.                                  | Switch to cash or wait for connection.             |
| Discount code invalid      | Code not found or expired                         | Red inline error under code field: 'Invalid discount code.'          | Try a different code or remove code.               |
| OTP too many attempts      | 3 wrong OTPs entered                              | Screen locked, countdown timer shown.                                | Wait for countdown, request new OTP.               |
| Barcode not found (POS)    | Scanned SKU not in products                       | Red toast: 'Product not found. Check SKU.'                           | Manual product search.                             |
| AI estimator — vague input | Gemini cannot determine material                  | Final card with specific guidance message.                           | Ask for more detail + retry.                       |
| Image upload failed        | File too large or wrong type                      | Red border on dropzone + 'File must be PNG/JPG/WebP under 10MB'.     | Choose a different image.                          |
| POS sync failure           | Background Sync POST fails                        | Orange badge: 'X sales failed to sync.'                              | Tap to view failed sales, retry manually.          |
| Stocktake lock conflict    | Someone else started a stocktake in this location | 'Error! A stocktake is already in progress at this location.'        | View in progress stocktake.                        |

## Part 10 — Mobile-Specific Considerations

### 10.1 Mobile Navigation Pattern

Customer store uses a bottom navigation bar on mobile (< 768px viewport width). This is the primary navigation surface on mobile — the header is reduced to logo + cart icon + language toggle only.

| Tab        | Icon         | Route           | Badge                         | Active State               |
|------------|--------------|-----------------|-------------------------------|----------------------------|
| Home       | House icon   | /               | None                          | Orange icon + orange label |
| Categories | Grid icon    | /categories     | None                          | Orange icon + orange label |
| Search     | Search icon  | /search         | None                          | Orange icon + orange label |
| Orders     | Package icon | /account/orders | Active orders count (if any)  | Orange icon + orange label |
| Account    | Person icon  | /account        | Notification dot if logged in | Orange icon + orange label |

**Bottom nav visibility rules:** Hidden when: virtual keyboard is open (browser resize event), user is in a full-screen modal or drawer, user is on /auth/\* pages, user is on /pos/\* pages. Shown again when keyboard closes (resize event fires with increased height).

### 10.2 Gesture & Touch Interactions

| Interaction           | Location                       | Behaviour                                                                         |
|-----------------------|--------------------------------|-----------------------------------------------------------------------------------|
| Swipe left/right      | Product detail image gallery   | Navigate between product images. Snaps to each image.                             |
| Swipe left            | Order history cards            | Reveals 'View Details' action button.                                             |
| Swipe down            | All lists (categories, orders) | Pull-to-refresh. Shows loading spinner at top. Fetches fresh data.                |
| Pinch-to-zoom         | Product detail primary image   | Zooms into product image. Double-tap resets zoom.                                 |
| Long press            | Cart items                     | Reveals inline delete option without needing to tap the × button.                 |
| Swipe left            | POS cart item                  | Reveals red 'Remove' action (confirms on swipe complete).                         |
| Swipe up              | Filter/sort bottom sheet       | Expands the filter panel on product listing.                                      |
| Back gesture (native) | Any page                       | Android system back + iOS swipe-from-left. Handled by Next.js router. Alert shown |

### 10.3 Keyboard & Input Handling

| Context              | inputmode / type Attribute  | Behaviour                                                                          |
|----------------------|-----------------------------|------------------------------------------------------------------------------------|
| Phone number input   | inputmode='tel', type='tel' | Numeric keyboard with +/# on iOS. Dial pad on Android.                             |
| OTP input (6 digits) | inputmode='numeric'         | Numeric keyboard. Auto-advance to next digit on entry. Paste from clipboard auto-f |
| PIN input (4 digits) | inputmode='numeric'         | Numeric keyboard. Dots shown, not digits. Paste disabled.                          |

| Context            | inputmode / type Attribute | Behaviour                                                                                  |
|--------------------|----------------------------|--------------------------------------------------------------------------------------------|
| Price input (SLSH) | inputmode='numeric'        | Numeric keyboard. No decimal allowed. Formatted with commas on blur.                       |
| Search             | inputmode='search'         | Shows 'Search' key on iOS. Submits on Enter/Search key.                                    |
| POS cash tender    | inputmode='decimal'        | Numeric with decimal point. Auto-select all on focus for fast re-entry.                    |
| Quantity stepper   | inputmode='numeric'        | Small inline numeric input. +/- buttons on either side. Tap field = full select for direct |

10.4 PWA Install Flow

| Step | Trigger / Screen        | Behaviour                                                                                                    |
|------|-------------------------|--------------------------------------------------------------------------------------------------------------|
| 1    | Deferred install prompt | Browser fires 'beforeinstallprompt'. Event is saved. App does NOT show browser's default banner immediately  |
| 2    | Custom install banner   | After user's 3rd visit to the customer store, show a custom banner at the bottom: 'Install Borama Hardware A |
| 3    | Install triggered       | On 'Install' tap, saved prompt.prompt() is called. Browser shows install dialog.                             |
| 4    | Post-install            | App opens in standalone mode (no browser chrome). Onboarding tooltip: 'You can now use the app offline!      |
| 5    | iOS Safari instruction  | iOS does not support beforeinstallprompt. On iOS Safari, show a separate banner: 'Tap Share → Add to Ho      |

## Part 11 — Transition & Animation Specifications

All animations respect prefers-reduced-motion: when set, animations are replaced with instant state changes or simple opacity fades (max 150ms). The animation system uses Tailwind CSS transitions + Framer Motion for complex sequences.

### 11.1 Page Transitions

| Transition                 | Type                        | Duration | Easing                         |
|----------------------------|-----------------------------|----------|--------------------------------|
| Page enter (Next.js route) | Fade in                     | 150ms    | ease-out                       |
| Page exit                  | Fade out                    | 100ms    | ease-in                        |
| Modal open                 | Scale 95%→100% + fade       | 200ms    | ease-out                       |
| Modal close                | Scale 100%→95% + fade       | 150ms    | ease-in                        |
| Drawer open (right)        | Slide from right            | 250ms    | cubic-bezier(0.32, 0.72, 0, 1) |
| Drawer open (bottom)       | Slide from bottom           | 300ms    | cubic-bezier(0.32, 0.72, 0, 1) |
| Drawer close               | Reverse slide               | 200ms    | ease-in                        |
| Toast appear               | Slide + fade from top right | 250ms    | ease-out                       |
| Toast dismiss              | Slide + fade out            | 200ms    | ease-in                        |

### 11.2 Component Animations

| Component             | Animation                                                                                         | Notes                                    |
|-----------------------|---------------------------------------------------------------------------------------------------|------------------------------------------|
| Add to Cart button    | Brief scale pulse: 100%→95%→100% over 120ms                                                       | Cart icon in header bounces once.        |
| Cart count badge      | Scale pop: 80%→120%→100% over 200ms on increment                                                  | Orange badge on cart icon.               |
| OTP digit input       | Smooth scale-in per digit as typed.                                                               | Each dot animates individually.          |
| Payment confirmed     | Green checkmark draws with stroke animation over 600ms                                            | Followed by subtle confetti burst.       |
| Tier upgrade modal    | Fullscreen confetti (canvas-confetti). Badge scales in 5 seconds, then auto-dismiss.              |                                          |
| Loyalty points earned | '+XXX pts' float-up animation from purchase button                                                | Appears once, fades in 1.5s.             |
| POS new order         | Card slides in from top with subtle shadow.                                                       | Accompanied by chime (if sound enabled). |
| Stock alert           | Alert card flashes amber border once on first appearance                                          | Non-looping.                             |
| Skeleton screens      | Shimmer animation — light sweep across grey blocks                                                | 1.5s loop, staggered per element.        |
| Offline banner appear | Slide down from above header.                                                                     | Auto-hides with reverse on reconnect.    |
| Chatbot open          | Panel scales from chat button origin point.                                                       | Framer Motion layout animation.          |
| Order status update   | Realtime: new status pill fades in, old one fades out                                             | 300ms crossfade.                         |
| Form validation error | Input border turns red with shake animation (3×). 150ms shake. Accessible: error text also shown. |                                          |

### 11.3 Print Styles

All receipt and report pages include a print stylesheet (@media print) for 80mm thermal printers used at POS counters.  
Print rules:

| Print Rule             | Behaviour                                                                                               |
|------------------------|---------------------------------------------------------------------------------------------------------|
| Hide navigation chrome | Header, sidebar, bottom nav, chat widget, toast container all hidden.                                   |
| Paper width            | 72mm content width (80mm paper - margins). Font size 10pt base.                                         |
| Receipt content        | Store name + address, date/time, cashier, order number barcode, items table, totals, payment method, ch |
| Page breaks            | Each receipt/report starts on a new page. No orphan headings.                                           |
| Barcode rendering      | react-barcode SVG renders correctly in print. No rasterization artifacts.                               |
| Colours                | Black and white only. No backgrounds. Orange brand colour converted to dark grey.                       |
| Font                   | System monospace (Courier) for receipt, system sans-serif for reports.                                  |

**Borama Hardware Store**  
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Borama, Somaliland

This document is confidential and intended solely for  
the development team. Version 1.0 — July 2025